

Harry Greateorex | PhD

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SKILLS

Communication: Talks and seminars at NASA Goddard Space Flight Center, ETH Zurich, Montana State University, Southwest Research Institute, and the SDO Workshop (Boulder); five peer-reviewed papers (three as first author) in *The Astrophysical Journal* and *Solar Physics*; teaching and outreach in schools and community groups.

Sales & Finance: Five years meeting individual and team sales targets at Millets; personal investor using Revolut's trading platform; daily follower of financial news and markets.

Project Delivery: Designed and led my own research projects from proposal through to publication; wrote a £250k funding proposal that scored above 80% at initial review.

Collaboration: Science-team member for the SoSpIM instrument on JAXA's Solar-C mission; contributor to ESA's FlaMEX mission proposal (shortlisted); member of an International Space Science Institute team; supervisor and mentor to research students since 2022.

Technical: Python, SQL, Git, LaTeX; statistical analysis of multi-terabyte satellite datasets; supervised machine-learning classification.

Organisation: Chaired a session and served on the Local Organising Committee at the Hinode/IRIS Joint Science Meeting (UCL, 2025); worked part-time at Millets for five years alongside full-time study.

ADDITIONAL EXPERIENCE

Public Engagement

Invited speaker to community astronomy groups; annual contributor to the NI Science Festival; outreach stand at the Girls in Maths & Physics event; sixth-form programming workshops under the QUB Pathway Opportunity Programme.

PROFILE

Physics PhD graduate (2025) working as a postdoctoral researcher in Belfast. I have over five years of experience in independent and collaborative scientific research, involving project conception, funding acquisition, data analysis, results delivery, and defence of findings. I have a wealth of experience communicating scientific research across all levels, from NASA mission teams and international conferences to school classrooms and community groups. Customer and client focussed in several capacities at every career stage. Now, an active personal investor with a strong interest in the financial industry, I am looking to build a commercial career centred on financial and technological innovation.

EXPERIENCE

Postdoctoral Research Fellow

2025–Present

Astrophysics Research Centre, Queen's University Belfast

- Led the largest statistical study to date of "EUV late-phase" solar flares, analysing 15 years of observations from NASA's Solar Dynamics Observatory covering several thousand events. The resulting paper, with me as first author, was accepted by *Solar Physics* in May 2026.
- Wrote and submitted a £250k research funding proposal, including the scientific case, budget, and delivery plan. The proposal scored above 80% at initial review.
- Presented my results in talks and seminars at NASA Goddard Space Flight Center, ETH Zurich, Montana State University, and Southwest Research Institute (San Antonio), and at international conferences. Through these I can explain difficult technical material to both specialist and non-specialist listeners.
- Chaired a session at the Hinode/IRIS Joint Science Meeting (UCL, 2025) and served on its Local Organising Committee.
- Science-team member for the SoSpIM instrument on JAXA's Solar-C mission and contributor to ESA's FlaMEX mission proposal, which was shortlisted. Member of an International Space Science Institute team studying how solar activity affects the Earth's upper atmosphere.
- Recruited and supervised a summer research intern, defining the project scope and milestones; their work contributed to a peer-reviewed publication. Have mentored MSci research students since 2022, all of whom achieved second-class honours or above.
- Manage my own workload and priorities across research, supervision, and collaboration commitments with minimal oversight.

Sales Assistant

2015–2020

Millets, Derby

- Met individual and team sales targets over five years in a KPI-driven store, working part-time alongside full-time study.
- Provided specialist advice on purpose-built technical equipment (footwear, waterproofs, camping and expedition kit), matching products to each customer's requirements and budget.

Team Coordination

Managed a recreational football team across multiple seasons — securing sponsorship and handling logistics, scheduling, and communications.

Workplace Inclusivity

Participated in workplace initiatives on diversity and equity, supporting an inclusive professional environment.

EDUCATION

Doctor of Philosophy (PhD) in Astrophysics 2021–2025

Queen's University Belfast

Thesis: “Observational Analysis of Lyman-Alpha Emission in Solar Flares”

- Secured competitive STFC funding for a four-month research placement at NASA Goddard Space Flight Center as a visiting scholar.
- Published a first-author paper (*Solar Physics*, 2024) identifying previously unrecognised discrepancies between competing satellite instruments, work that required detailed knowledge of each instrument's design and calibration.
- Served as a Chief Observer in the Max Millennium solar-flare forecasting programme since 2023, combining data from multiple observatories into daily forecasts issued to a global mailing list.

MSci in Physics (First Class Honours)

2017–2021

Queen's University Belfast

Final-Year Project: “Computational Modelling of the Chromospheric Response to Solar Flare Heating”

- Specialised in astrophysics with additional coursework in plasma physics, medical physics, and materials science.

PUBLICATIONS

Greatorex, H. J., et al. (2026). “The EUV Late Phase: Statistical Results from 15 Years of Solar Dynamics Observatory Observations.” *Solar Physics* (accepted).

Greatorex, H. J., Milligan, R. O., & Dammasch, I. E. (2024). “On the Instrumental Discrepancies in Lyman-Alpha Observations of Solar Flares.” *Solar Physics*, 299, 162.

Greatorex, H. J., & Milligan, R. O. (2023). “Observational Analysis of Ly α Emission in Equivalent-Magnitude Solar Flares.” *The Astrophysical Journal*, 954, 120.

Two further co-authored publications; a sixth paper in preparation.